

DocMind — Public Demo Test Document

A self-contained document for demonstrating PDF question answering, retrieval, multi-hop reasoning, and numerical extraction.

1. Project Overview

DocMind is a Retrieval-Augmented Generation (RAG) document assistant designed to answer questions from uploaded documents. For the public demo, a pre-indexed demonstration document is used so visitors can ask questions without uploading or processing a new PDF on the hosted backend.

2. Demonstration Project: Aurora Analytics

Aurora Analytics conducted a six-week evaluation of its internal document-search assistant. The evaluation compared three retrieval configurations: Baseline Search, Hybrid Retrieval, and Hybrid Retrieval with Reranking. The goal was to determine whether combining keyword and semantic retrieval could improve answer quality while keeping response latency acceptable.

3. Evaluation Setup

The evaluation used 1,200 questions created from a collection of 300 internal documents. Each configuration was tested against the same question set. The team measured retrieval recall, answer accuracy, and median response latency. A successful answer was defined as one that contained the required information and did not contradict the source document.

Configuration	Retrieval Recall	Answer Accuracy	Median Latency
Baseline Search	71%	68%	1.4 s
Hybrid Retrieval	84%	79%	1.8 s
Hybrid + Reranking	91%	87%	2.3 s

4. Key Findings

- Hybrid Retrieval improved retrieval recall from 71% to 84%, a gain of 13 percentage points over Baseline Search.
- Hybrid Retrieval with Reranking achieved the highest retrieval recall at 91% and the highest answer accuracy at 87%.
- The reranked configuration increased median latency to 2.3 seconds, which was 0.5 seconds slower than Hybrid Retrieval without reranking.
- The team considered 2.5 seconds an acceptable median latency for the internal use case.
- The evaluation suggested that the additional latency from reranking was justified by the improvement in answer quality.

5. Implementation Notes

The demonstration system stores pre-processed document chunks and their vector representations before deployment. During a user query, the system retrieves the most relevant chunks and passes them to the language model as context. This reduces server-side work during the public demo because PDF parsing, chunking, and indexing do not happen for every visitor.

6. Conclusion

For the Aurora Analytics evaluation, Hybrid Retrieval with Reranking provided the strongest overall retrieval and answer-quality results. Although it introduced additional latency, the 2.3-second median remained within the team's 2.5-second target. The results support using reranking when answer quality is more important than minimizing latency.